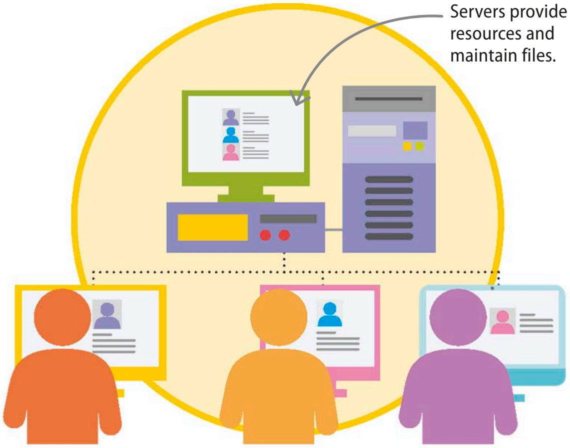
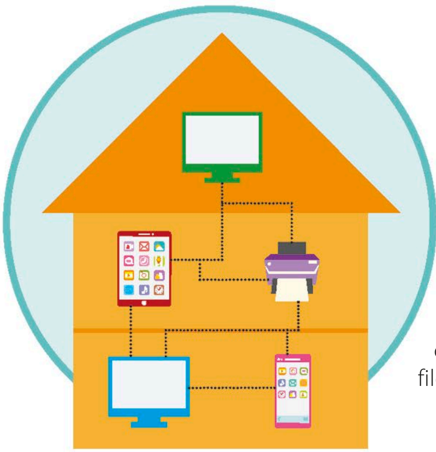


Client-server networks

An ordinary computer – or "client" – is designed to interact with users for tasks such as text editing, photo browsing, or video streaming. A server, on the other hand, is designed to interact with clients. Servers run specialized software, and have specialized operating systems. They can be used to host websites and databases.

◁ **Client-server model in a school**
In a school network, grades are stored on a central computer called a server. Each teacher's personal computer can connect to this main server to access the grades and make changes.



▷ **Home network**
Home networks are generally P2P. One computer might be connected to a printer, another to a scanner, while a third is connected to the TV and stores all the videos.

Peer-to-peer networks

A peer-to-peer (P2P) network has no specialized servers. Instead, each computer alternates between taking the role of client and server. P2P networks are easy to set up, but hard to maintain because each device needs to be constantly operational. If a computer crashes, all its resources and files are cut off.

Using a network

Sharing resources is a good way to save money – whether it's sharing software, data, or access to hardware. Imagine the hassle if every computer in a building needed its own printer. Unfortunately, networks come with security risks. It's easy enough to protect a network from outsiders, but once inside it can be difficult to set up barriers.

Advantages	Disadvantages
It is easier to collaborate and communicate through a network.	Networks cost money. The more complicated it is, the more expensive a network becomes to set up.
Documents are stored in a central location where everyone can access them.	Networks require constant troubleshooting, updating software, and management.
Since there's a single copy of each file, versions can't get out of synchronization.	If the central hub breaks down, the entire network is disrupted.
Entry to the network can be restricted, and access to files can be controlled.	Viruses and malware can spread more easily through a network.
It's easy to ensure that important data is properly saved and backed up.	Streaming eats up bandwidth, so a single device can slow down the network for everybody.